SUPPORTED BY

MINISTRY OF POWER
GOVERNMENT OF INDIA

NITI Aayog

ENERGY DEPARTMENT
Government of Rajasthan

MINISTRY OF POWER
GOVERNMENT OF INDIA
CENTRAL ELECTRICITY AUTHORITY

AIDA
Collaboration | Innovation | Excellence

DISTRIBUTION UTILITY MEET
DUM 2026

ORGANIZER
ISGF
India Smart Grid Forum

KNOWLEDGE PARTNER
Power Foundation of India

HOST UTILITIES
JVVNL
Jodhpur Vidyut Vitran Nigam Limited
Ajmer Vidyut Vitran Nigam Limited

POWERED BY
Pelocal

RE PARTNER
RREC

CO - HOST UTILITIES
TATA POWER
Chandigarh Power Distribution Limited (CPDL)
TATA POWER-DDL
BSES Rajdhani Power Limited
NPCL
BSES Yamuna Power Limited

DUM 2026 CONFERENCE SESSION

SESSION 5: CLIMATE RESILIENCE AND SECURING THE PHYSICAL ASSETS OF DISCOMs

Pricing Disruption to Justify Resilient Distribution | Lessons from India, Chile and beyond

Presented By

KIRAN SABBINENI, LEAD SPECIALIST – ENERGY, CDRI

Distribution Utility Meet | 27 - 28 October 2026 | www.dumindia.in

EVOLUTION OF CDRI's RISK ASSESSMENTS

DISTRIBUTION UTILITY MEET DUM 2026

ISGF India Smart Grid Forum

ODISHA, INDIA · 2021–23
Cyclone and flood resilience of state T&D

CHILE · 2026–27
National study across the power supply chain

NEXT
Every DISCOM, using its own data

200,000 +

11 kV poles downed by Fani (2019): built for ~150–180 km/h, hit by 205 km/h

$ 1.2 B Resilience investment

22 hazards

screened across 180,677 km of distribution lines

Today and under future climate scenarios

Digital grid

GIS, smart-meter and outage data can scale the method nationally

Risk-informed capex, not one-off studies

Distribution is the largest, most exposed asset base – and the hardest one to justify investing in.

LESSON 1: BETTER DATA, SHARPER DECISIONS

DISTRIBUTION UTILITY MEET DUM 2026

ISGF India Smart Grid Forum

Insights gets sharper as the data improves

As utility data goes digital, the next step comes within reach

ODISHA · 2021

- Proved the method with data of the day
- Substation locations as the asset layer
- Historical hazard maps

CHILE · 2026

- Open national GIS: 180,677 km of distribution lines
- Climate atlas with future scenarios
- Outage causes recorded by cause code, locality, projections

NEXT

- Asset GIS down to transformers and poles
- Asset age and replacement value in the registry
- Outage records tagged with cause and asset
- Climate projections turned into design values

The opportunity: as DISCOM data goes digital, capture asset age, value and outage cause – and every DISCOM can price its climate risk.

LESSON 2: WHAT HAZARD MAPS MISS

5–25%

of outage minutes (SAIDI) caused by weather, by region – Chile, 2019–25

~750 h

longest outage, caused by a wildfire. Most weather outages last under 10 hours

“Low”

wind-map rating for distribution lines – yet fallen trees are a top outage cause

HEAT IS THE RISING HAZARD

In Chile's future climate scenarios, a significant share of distribution lines moves to “Very High” heat exposure.

India today: ~1.3 million distribution transformer failures a year.

WHAT DISCOMs CAN DO

- Check hazard maps against your own outage-cause data
- Run vegetation risk as its own track
- Plan restoration for the long tail, not the average
- Rate transformers and conductors for future heat

Source: CDRI–Arup Power Sector Resilience Study in Chile, Hazard & Exposure Assessment (preliminary, Oct 2026); DUM 2026 Session 5 brief.

LESSON 3: PRIORITISE BY WHO LOSES POWER

1 EXPOSED ASSETS
Lines, substations and transformers in harm's way

2 CRITICAL CUSTOMERS SERVED
Hospitals, water, telecom – Chile uses the legal list

3 WHERE TO ACT FIRST
Hardening, microgrids and first restoration crews

FAST-TRACK URGENT WORKS
Chile's Law 21.721 lets “necessary and urgent works” skip the ordinary planning cycle.

MONEY READY BEFORE THE EVENT
A 1.5%-of-revenue disaster reserve, parametric insurance, contingent credit.

UTILITY LIABILITY IS RISING
A Chilean prosecutor held utilities responsible for the 2023 Biobío wildfires.

Rank investment by the consequence of failure, not only by the hazard at the asset.

PRICE THE DISRUPTION, NOT JUST THE DAMAGE

DISTRIBUTION UTILITY MEET DUM 2026

ISGF India Smart Grid Forum

WHAT AN OUTAGE REALLY COSTS

Utility asset repair, lost revenue — ✔ ON THE BOOKS TODAY

Customers lost output, backup diesel — RARELY PRICED ▼

Critical services hospitals, water, telecom, transport

Economy supply chains, jobs, public health

~US$390 bn a year: cost of infrastructure disruptions to firms and households in LMICs (World Bank, *Lifelines*)

PRICE IT ACROSS THE WHOLE EVENT

ABSORB	RESPOND	RECOVER
MEASURE: Customers and critical loads kept on	MEASURE: Hours to restore hospitals, water, telecom	MEASURE: Customer-hours lost over the full event
PRICE: Damage and outage avoided	PRICE: Hourly cost to critical services	PRICE: Cost that rises with duration

GOVERNMENTS ALREADY PRICING IT – their regulators can justify resilience capex on its value to society

UK · Ofgem
Outage minutes priced into network revenue; storm compensation

AUSTRALIA · AER
Value of Customer Reliability, plus a Value of Network Resilience for long outages

USA · DOE / LBNL
ICE Calculator puts a value on outage cost in reliability and resilience cases

CHILE · CDRI study
Multi-sector disruption costing with Ministry of Energy; results 2027

ASKS TO REGULATORS

- **Value of lost load and of resilience**
- **Approve capex on its benefit to society**
- **Build utility data to feed it**

Count what an outage costs everyone, for as long as it lasts – and resilient distribution pays for itself.

THANK YOU

For discussions/suggestions/queries email: ***dum@indiasmartgrid.org***

www.dumindia.in

Kiran Sabbineni | kiran.sabbineni@cdri.world | www.cdri.world

